

HAMILTON–BOLTZMANN COTANGENT GEOMETRY FROM DETERMINISTIC HARD-SPHERE COLLISIONS

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ABSTRACT. The raw perspective-entropy Hessian may be indefinite. On a regular short-time constrained phase we add the KKT multiplier curvature and prove coercivity from an explicit linearized kinetic energy estimate. Finite-volume marked cumulants construct the Gaussian covariance first; backward observability identifies the complete balance gauge, and second-order Mosco convergence identifies the covariance with the closed constrained tangent form.

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1. ROUND-NINE RECONSTRUCTION: CONSTRAINED CURVATURE, COVARIANCE-FIRST TANGENTS, AND KINETIC OBSERVABILITY

The raw second variation of the perspective entropy is not asserted to be positive. Positivity is obtained only on a regular, sufficiently short constrained phase by combining the dynamic Hessian with the multiplier curvature dictated by the linearized balance. Independently, finite-volume cumulants first construct the Gaussian covariance. The variational quadratic

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form is then identified with that covariance by a second-order epi-convergence theorem, avoiding the circular definition of one through the other.

1.1. Weighted tangent and cotangent spaces. Fix $m > 8$. Let X_T be the completion of smooth density tangents for

$$\|u\|_{\mathsf{X}_T}^2 = \sup_{t \leq T} \int \frac{|u_t|^2}{f_t} (1 + |v|^m) + \int_0^T \|u_t\|_{H_{x,v}^{-1}(w_m)}^2 dt,$$

and let Y_T be the weighted $L^2((qA_f)^{-1})$ space of contact current tangents. The closed linearized balance operator is

$$\mathcal{B}(u, J) = \partial_t u + v \cdot \nabla_x u - \Delta^* J.$$

Together with its endpoint and conservation coordinates. Its kernel with the microcanonical initial tangent condition is denoted $\mathsf{T}_{f,q}$. The cotangent pair (p, ψ) is taken in the corresponding weighted transport Sobolev/Orlicz space and acts through $\Delta p + \psi$.

For a regular biased path (f, qA_f) the exact unconstrained second variation is

$$\begin{aligned} \mathfrak{h}_{f,q}(u, J) &= D^2 I_0(f_0)[u_0, u_0] + \int_0^T \int \frac{|J - qDA_f[u]|^2}{q} dA_f dt \\ &\quad + \int_0^T \int (1 - q) D^2 A_f[u, u] dt. \end{aligned}$$

The last term can have either sign.

1.2. The KKT-completed tangent curvature. Let p^* be the regular Lagrange multiplier for the weak balance at the biased phase. The constrained Lagrangian is

$$\mathcal{L}(f, \Gamma, p) = I_0(f_0) + \int \ell(d\Gamma/dA_f) dA_f + \langle p, \mathcal{B}(f, \Gamma) \rangle.$$

Its second variation in (f, Γ) contains the additional multiplier curvature

$$\mathfrak{c}_{p^*}(u) = \int_0^T \int qA_f D_f^2(\Delta p^*)[u, u] dt + \text{endpoint preparation curvature}.$$

For the binary Boltzmann reference the first term is the polarization of the bilinear collision intensity evaluated against the biased collision increment.

Lemma 1.1 (Linearized kinetic energy estimate). *On every compact regular phase chart there are C_0, C_1 such that every $(u, J) \in \mathsf{T}_{f,q}$ satisfies*

$$\|u\|_{\mathsf{X}_T}^2 \leq C_0 \|u_0\|_{L^2(f_0^{-1}w_m)}^2 + C_1 \int_0^T \int \frac{|J - qDA_f[u]|^2}{q} dA_f dt.$$

The constants are uniform for $T \leq T_$. Moreover*

$$|\mathfrak{h}_{f,q}^{\text{neg}}(u) + \mathfrak{c}_{p^*}^{\text{neg}}(u)| \leq CT \|q - 1\|_{L_w^\infty} \|u\|_{\mathsf{X}_T}^2.$$

Proof. Test the linearized balance with the solution of the backward weighted transport–collision equation having datum u/f . Free transport is skew-adjoint after the endpoint terms, while the symmetric part of the linearized collision operator controls the orthogonal complement of the collision invariants. The finite-dimensional invariant modes are fixed by the initial micro-canonical tangent condition. Cauchy–Schwarz gives the contact residual term and Gronwall gives the first estimate. The bilinearity of A_f , boundedness of the regular multiplier, and the velocity moments give the second estimate. All boundary pairings are justified by the B2 closed Green graph. $\square$

Theorem 1.2 (Positive constrained curvature). *Choose the regular horizon so that*

$$CT_* \|q - 1\|_{L_w^\infty} < \frac{1}{2}.$$

Then the KKT-completed form

$$\mathfrak{q}_{f,q}(u, J) = \mathfrak{h}_{f,q}(u, J) + \mathfrak{c}_{p^*}(u)$$

is continuous and coercive on $\mathsf{T}_{f,q}$:

$$\mathfrak{q}_{f,q}(u, J) \geq c \left(\|u_0\|_{L^2(f_0^{-1}w_m)}^2 + \|u\|_{\mathsf{X}_T}^2 + \int \frac{|J - qDA_f[u]|^2}{q} dA_f dt \right).$$

No positivity is claimed for the raw perspective Hessian off the constrained tangent or outside the regular short-time phase chart.

Proof. The positive initial entropy and square contact residual control the right side of Lemma 1.1. Absorb the combined negative curvature using the displayed small-horizon condition. The remaining KKT terms are bounded symmetric perturbations, proving continuity and coercivity. $\square$

1.3. Closed range and the complete gauge. Let $\mathcal{B}^*$ be the backward kinetic adjoint. Constants and collision invariants form a finite-dimensional space $\mathcal{N}$; fix their endpoint normalization.

Theorem 1.3 (Backward observability and closed balance range). *For every normalized cotangent (p, ψ) ,*

$$\|(p, \psi)\|_{\mathsf{P}_T/\mathcal{N}} \leq C \left(\|\mathcal{B}^*(p, \psi)\|_{\mathsf{T}_{f,q}^*} + \|\Delta p + \psi\|_{L^2(qA_f)} \right).$$

Consequently $\mathcal{B}$ has closed range and

$$\ker \mathfrak{D} = \{(r, -\Delta r) : r \in \mathsf{P}_T\}, \quad \mathfrak{D}(p, \psi) = \Delta p + \psi.$$

Thus the identifiable cotangent is the Hausdorff quotient by the full balance gauge, not merely by collision invariants.

Proof. Solve the backward equation blockwise by Duhamel around the free-transport characteristics. The symmetric collision part has a spectral gap off $\mathcal{N}$; the macroscopic modes satisfy a finite hyperbolic system whose endpoint data are controlled by the balance functional. Velocity truncation and the B2 trace estimate pass the result to the weighted spaces. This gives the

observability inequality. The Hilbert closed-range theorem applies to $\mathcal{B}$. If $\mathfrak{D}(p, \psi) = 0$, the dual action on every balanced tangent vanishes; closed range and the adjoint characterization give $(p, \psi) = (r, -\Delta r)$ after the invariant normalization. The converse is the exact weak balance identity. $\square$

1.4. Finite-volume covariance first. Let Z_ε be the exact-centred joint density/contact fluctuation field under a regular finite-volume source:

$$Z_\varepsilon = \sqrt{\mu_\varepsilon} ((\pi^\varepsilon, \Gamma^\varepsilon) - DQ_\varepsilon(\Theta)).$$

For a finite test family $F = (F_1, \dots, F_d)$ set $\Sigma_\varepsilon(F) = D^2Q_\varepsilon(\Theta)[F, F]$.

Theorem 1.4 (Finite-dimensional covariance and Gaussian limit). *For every finite weighted test family,*

$$\Sigma_\varepsilon(F) \longrightarrow \Sigma(F),$$

and the characteristic function of $F(Z_\varepsilon)$ converges to that of $N(0, \Sigma(F))$. The family $(\Sigma(F))$ is projectively compatible and positive semidefinite. Its null space consists precisely of the balance gauge and the prepared conservation directions.

Proof. Normal convergence of the B2 complex marked pressure gives convergence of all finite derivatives. The k th cumulant of the scaled field is $O(\mu_\varepsilon^{1-k/2})$ for $k \geq 3$, while the second converges to the Hessian. This proves the Gaussian characteristic function. Positivity is inherited from finite covariance matrices. A zero-variance class is constant under every local activity/contact perturbation. Varying one free-flight cell and one actual binary contact shows that it annihilates every balanced tangent; Theorem 1.3 identifies it with the full gauge plus prepared constraints. $\square$

The compatible finite-dimensional covariances define a Gaussian cylindrical measure on the dual of the nuclear test core. They do not imply a bounded inverse on the whole Hilbert completion.

1.5. Second-order epi convergence. Let I_ε be the finite-volume convex rate on a finite projection, let $x_\varepsilon = DQ_\varepsilon(\Theta)$, and define

$$J_\varepsilon(z) = \mu_\varepsilon [I_\varepsilon(x_\varepsilon + z/\sqrt{\mu_\varepsilon}) - I_\varepsilon(x_\varepsilon) - \Theta \cdot z/\sqrt{\mu_\varepsilon}].$$

Theorem 1.5 (Mosco identification of the tangent action). *On every finite projection, J_ε Mosco-converges to $\frac{1}{2}\Sigma^{-1}$ on the covariance range. Passing through the increasing balanced test core yields the closed quadratic form associated with $\mathfrak{q}_{f,q}$ from Theorem 1.2. Hence the Lax–Milgram inverse of the KKT-completed form and the covariance constructed in Theorem 1.4 agree on their common form domain.*

Proof. On a finite projection, analytic Taylor expansion of Q_ε and Legendre duality give the quadratic liminf. For a vector in the covariance range, tilt by $\Theta + h/\sqrt{\mu_\varepsilon}$ to obtain a recovery sequence and the matching limsup. Projectively increasing the finite test spaces gives a monotone sequence

of closed forms. The balance constraint and B1 initial Schur complement are continuous on the form domain. The explicit second variation of the constrained Lagrangian is exactly $\mathbf{q}_{f,q}$, so uniqueness of the Mosco limit identifies the two forms. Coercivity on the regular chart then gives the Lax–Milgram representation. $\square$

1.6. Process tightness from localized cumulants. For a time interval I and a test F , let $Z_\varepsilon(I, F)$ be the increment of the exact-centred field. The B2 source expansion localized to deterministic intervals gives connected cumulant densities with one anchor time and tree-decaying relative times.

Lemma 1.6 (Fourth-moment increment estimate). *For every finite test family there are $C, \eta > 0$ such that*

$$\mathbb{E}|Z_\varepsilon([s, t], F)|^4 \leq C|t - s|^2 + C\mu_\varepsilon^{-1}|t - s|,$$

and

$$\mathbb{E}|Z_\varepsilon([s, t], F)|^2 \leq C|t - s|.$$

The bounds include intervals sharing an endpoint; no product $|I||J|$ estimate is assumed without a cumulant-density integration.

Proof. Differentiate the B2 pressure with four independently supported time sources. A connected tree has one anchor time and each additional relative event time is integrated against the renewal kernel. Its L^1 norm is summable on the short horizon. Integrating the four-point cumulant density over $[s, t]^4$ gives $C|t - s|^2$; diagonal coincidences contribute the finite-size term $C\mu_\varepsilon^{-1}|t - s|$. The two-point statement is identical. Boundary atoms are already part of the trace-source cumulant and need no stopping-time substitution. $\square$

Theorem 1.7 (Prepared joint Gaussian process). *The density/contact fluctuation fields converge in the weighted distribution Skorokhod space to the unique centred Gaussian process with covariance Σ and linearized biased balance. The initial covariance is the B1 prepared Schur complement. Its Cameron–Martin form is the closed KKT-completed form of Theorem 1.5.*

Proof. Finite-dimensional convergence is Theorem 1.4. Lemma 1.6 gives Kolmogorov/Aldous bounds for each member of a countable nuclear test core; Mitoma’s criterion gives process tightness. Passing the exact microscopic Green balance to the limit identifies the linearized martingale problem. The covariance and initial law determine its unique Gaussian solution. Theorem 1.5 identifies the Cameron–Martin form. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Every cross-paper input is typed in the round-nine

dependency ledger. Repository proof and build gates are internal checks and do not replace independent external mathematical review.

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